1️⃣ Competency 7: Simplifying and decomposing a real-world problem (Highest relevance)

Definition:

> Simplify or decompose a situation or problem... formulate

🔍 How it appears in the task

The problem presents a complex real-world situation:

- A revolving door + airflow issue (whether air can pass through)
- A dynamic process (door rotating in different positions)

To solve it, students must simplify it into:

- A **static geometric problem** (circle + sectors + arcs)
- Key transformation:
  - "Air can flow through" ⇨ "two openings form a continuous path"

📌 Core insight

Students must perform a crucial modeling step:

- Ignore the dynamic rotation
- Focus on the **critical extreme case** (maximum opening that still prevents airflow)

👉 This is a typical case of:

> "Complex physical situation → simplified geometric model"

✔ This is the most central *formulate* competency in this task

2️⃣ Competency 12: Identifying constraints, assumptions, and simplifications

Definition:

> Identify the constraints, assumptions, simplifications formulate

🔍 How it appears in the task

Key constraints:

- Three wings always divide the circle equally
- Two openings have equal arc length
- Air must **not flow freely between entrance and exit**

Implicit assumptions:

- Only the **worst-case (maximum opening)** needs to be considered
- Airflow can be represented by geometric connectivity

📌 Key idea

Students must recognize:

- The task is not to find any arc length
- But to find a **maximum allowable value (a boundary condition)**

👉 This reflects:

> "Constraint identification and boundary analysis in modeling"

3️⃣ Competency 5: Recognising mathematical structure

Definition:

> Recognise mathematical structure... formulate

🔍 How it appears in the task

Students need to identify:

- Circular structure
- Three equal sectors (120° each)
- Openings correspond to arcs

📌 Transformation

Real-world situation ⇨ mathematical structure:

- Airflow problem ⇨ geometric coverage/connectivity problem
- Maximum arc length ⇨ constraint on central angle

👉 This reflects:

> "Structure recognition driving the modeling process"

4️⃣ Competency 4: Understanding and constructing a model

Definition:

> Read, decode... to create a model of the situation formulate

🔍 How it appears in the task

Students must interpret:

- Dotted arcs = door openings
- The diagram on the right = extreme case of airflow

📌 Key modeling steps

- Read the diagram → understand airflow path
- Construct:
  - "If arcs are too large ⇒ openings connect"

👉 This reflects:

> "From contextual understanding to model construction"

5️⃣ Competency 6: Identifying mathematical aspects of a real-world situation

Definition:

> Identify and describe the mathematical aspects... formulate

🔍 How it appears in the task

Extract key variables:

- Radius (from diameter 200 cm)
- Central angle (120°)
- Arc length (target variable)

📌 Core idea

Students must extract:

- Geometric variables
- Geometric relationships

6️⃣ Competency 2: Identifying key variables

Definition:

> Identify the key variables in a model formulate

🔍 Variables include:

- Circle radius
- Arc length of openings
- Door wing angles

7️⃣ Competency 9: Translating into standard mathematical representation

Definition:

> Translate a problem into a standard mathematical representation formulate

🔍 How it appears in the task

The situation is translated into:

- Arc length formula: $s = r\theta$
- Finding the maximum allowable θ

8️⃣ Competency 1: Selecting a mathematical representation

Definition:

> Select a mathematical description or representation formulate

🔍 How it appears in the task

Students choose:

- Circle model
- Sector (arc) model

1️⃣ 能力7：对现实问题进行简化与分解（最高相关）

对应条目：

> Simplify or decompose a situation or problem... formulate

🔍 题目体现

题目本质是一个复杂情境：

- 旋转门 + 气流问题（空气是否贯通）
- 门的动态过程（不同位置）

但求解时必须简化为：

- **静态几何问题（圆 + 扇形 + 圆弧）**
- 关键转化：
  - “空气可以通过” ⇨ “两个开口连通形成连续通道”

📌 本质分析

学生需要完成关键建模：

- 忽略动态过程（旋转过程）
- 聚焦最极端情况（最大开口仍不连通）

👉 属于典型：

> “复杂物理情境 → 简化几何模型”

✔ 这是本题最核心的 formulate 能力

2️⃣ 能力12：识别约束、假设与简化

对应条目：

> Identify the constraints, assumptions simplifications formulate

🔍 题目体现

关键约束：

- 三个门翼始终等分圆
- 两个开口弧长相等
- “不能形成空气自由流通”

隐含假设：

- 只需考虑“最不利情况”（最大开口）
- 空气通道可以用几何连通表示

📌 关键点

学生必须意识到：

- 不是求任意弧长
- 而是求**临界最大值（边界条件）**

👉 属于：

> “建模中的约束识别与边界分析”

3️⃣ 能力5：识别数学结构

对应条目：

> Recognise mathematical structure… formulate

🔍 题目体现

需要识别：

- 圆结构
- 三等分（120°）
- 开口 = 圆弧

📌 转化过程

现实问题 ⇨ 数学结构：

- 气流问题 ⇨ 几何覆盖问题
- 最大弧长 ⇨ 中心角约束

👉 属于：

> “结构识别驱动建模”

4️⃣ 能力4：理解并构建问题模型

对应条目：

> Read, decode… to create a model of the situation formulate

🔍 题目体现

学生需理解：

- 图中虚线弧表示“开口”
- 右侧图表示“空气贯通的极端情况”

📌 关键建模步骤

- 读图 → 理解气流路径
- 构建：
  - “弧长过大 ⇒ 连通”

👉 属于：

> “情境理解 → 模型构建”

5️⃣ 能力6：识别现实问题中的数学要素

对应条目：

> Identify and describe the mathematical aspects... formulate

🔍 题目体现

提取关键变量：

- 半径（由直径200 cm得到）
- 中心角（120°）
- 弧长（目标变量）

📌 本质

从复杂描述中抽取：

- 几何变量
- 几何关系

6️⃣ 能力2：识别关键变量

对应条目：

> Identify the key variables in a model formulate

🔍 变量包括：

- 圆半径
- 开口弧长
- 门翼角度

7️⃣ 能力9：转化为标准数学表示

对应条目：

> Translate a problem into a standard mathematical representation formulate

🔍 体现

将问题转化为：

- 弧长 = $r\theta$
- θ 的最大允许值

8️⃣ 能力1：选择数学表示

对应条目：

> Select a mathematical description or representation formulate

🔍 体现

选择：

- 圆模型
- 扇形模型